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CONFIDENCE IN YOUR ENVIRONMENTAL DNA STUDY

Regulatory acceptance of eDNA technology requires that results be both reliable and defensible. This can be achieved through proper field sampling, sample handling, laboratory assay design and validation.

There are a number of field and laboratory approaches to eDNA being practiced around the globe. While no standards exist for eDNA sampling and testing at this time, following available guidelines and best practices helps to ensure a successful eDNA study with accurate and dependable results.

IT ALL STARTS IN THE FIELD

A successful eDNA study begins with a survey design that is based on the objectives of the study in combination with knowledge of the target species' life history and habitat use.

DNA is not evenly spread throughout the environment, so the following survey design elements need to be considered:

- Habitat preference
- The time of year the species occupies a particular geography
- When the species is most active
- The lifecycle stage that introduces the most DNA from the species to the environment

Random environmental sampling without consideration of this information is unlikely to provide meaningful and reliable eDNA results.

Hobbs and Goldberg¹ provide an overview of considerations for eDNA sampling in their *Environmental DNA Protocol for Freshwater Aquatic Ecosystems* document, prepared for the BC Ministry of Environment in 2017. In this document, they present recommendations for eDNA study design, water sampling and filtration based on the results of scientific publications and their own research.

WHERE AND WHEN TO SAMPLE

Location and timing of eDNA sampling will depend on the habitat preference and life history of the target species. Field sampling should occur during times when the species is most active and sheds the most DNA in the environment. For most species, this coincides with the breeding season.

¹ Hobbs, J. and C. Goldberg. 2017. Environmental DNA Protocol for Freshwater Aquatic Ecosystems (Version 2.2). Prepared for: BC Ministry of Environment.



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The specific objectives of the study will determine the geographical coverage required and the amount of environmental sampling involved. Stagnant waters do not transport eDNA far from the source, whereas flowing water will move eDNA rapidly away from its source. To improve the probability of detection, it is also important to sample the specific micro-habitats the target species prefers.

Increasing the number of samples taken at each site improves the chance of capturing eDNA from the target species. Three water sample replicates of 1L volume are recommended for geographies where species distribution is unknown¹. Sampling across different seasons and across different years will provide greater information on the target species occupancy of the sampled geography and improves the probability of detection.

Water samples should be taken near the surface to avoid sediment and debris that can collect in the filter. Debris would otherwise slow the process of filtration or prevent filtering of the entire sample volume while also reducing the amount of eDNA collected and the ability to detect the target species.

GOOD FIELD PRACTICES

Good field practices are required to avoid contamination or carryover (on footwear, waders, boat hulls, reused sampling supplies, etc.) that can lead to false positives. Any bleach that is used to clean sampling and filtration supplies must be thoroughly and completely removed by rinsing with clean water before reuse. Otherwise, there is considerable risk of destroying the very eDNA you are sampling, resulting in false negatives.

Negative control samples should be included in the field and filtering steps to assess for possible contamination leading to false positives. An appropriate field negative control is a clean, ultra-pure water sample brought into the field and filtered at the same time and location as the collected field samples.

ROBUST ENVIRONMENTAL DNA LABORATORY ASSAY DESIGN

Confidence in eDNA test results requires a properly designed and validated laboratory assay. DNA from the target species is typically present at low concentrations in the environment and in various degrees of degradation in the collected eDNA sample. A robust, sensitive and specific laboratory assay is required to successfully detect only the target species DNA and at very low concentrations.

eDNA assays are typically designed to target either mitochondrial DNA (mtDNA) in animals or chloroplast DNA



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(ctDNA) in plants and algae. This is to increase chances of collecting and detecting the target species as there are many copies of mtDNA and ctDNA per cell as opposed to only one copy of nuclear DNA per cell.

There are three stages of eDNA assay development²:

Step 1: In Silico – assay design of qPCR primers and probes with bioinformatics software and evaluation for specificity to the target species through interrogation of all known DNA sequences available in the public database;

Step 2: In Vitro – assay is tested on high quality DNA obtained from a sample confirmed to be from the target species. Specificity is determined by testing against DNA from closely related species and human DNA. Sensitivity is evaluated through a DNA dilution series;

Step 3: In Situ – assay is tested on field samples from known positive and negative sites to evaluate for robustness in real world samples. This is an important step to ensure that the eDNA assay is fit for purpose.

LABORATORY ASSAY CONTROLS

The laboratory assay must include proper controls to monitor for type 1 (false positive) and type 2 (false negative) errors.

1. Check for DNA integrity – it is important to first check that the eDNA collected is of sufficient quality to control for false negatives. Without this step there is no way to distinguish whether a negative eDNA result is due to absence of the target DNA, assay inhibition, or simply due to the eDNA being of poor quality. Bureau Veritas has licensed a the IntegritE-DNA^{TM3} method from UVic Industry Partnerships, a technique developed by Dr. Caren Helbing.
2. Check for assay performance – positive control added to monitor assay performance and reliability;
3. Check for contamination – negative control (blank) added at eDNA extraction and qPCR stages to check for

² Herder, J.; Valentin, A.; Bellamain, E.; Dejean, T.; van Delft, J.J.C.W.; Thomsen, P.F.; Taberlet, P. 2014. Environmental DNA: a review of the possible applications for the detection of (invasive) species. Nijmegen: Netherlands Food and Consumer Product Safety Authority.

³ Hobbs, J.; Round, J.M.; Allison, M.J.; Helbing, C.C. 2019. Expansion of the known distribution of the coastal tailed frog, *Ascaphus truei*, in British Columbia, Canada, using robust eDNA detection methods. PLoS ONE 14(3):1-23



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contamination of the target DNA and control for false positives.

ENVIRONMENTAL DNA ASSAY SAMPLE REPLICATES

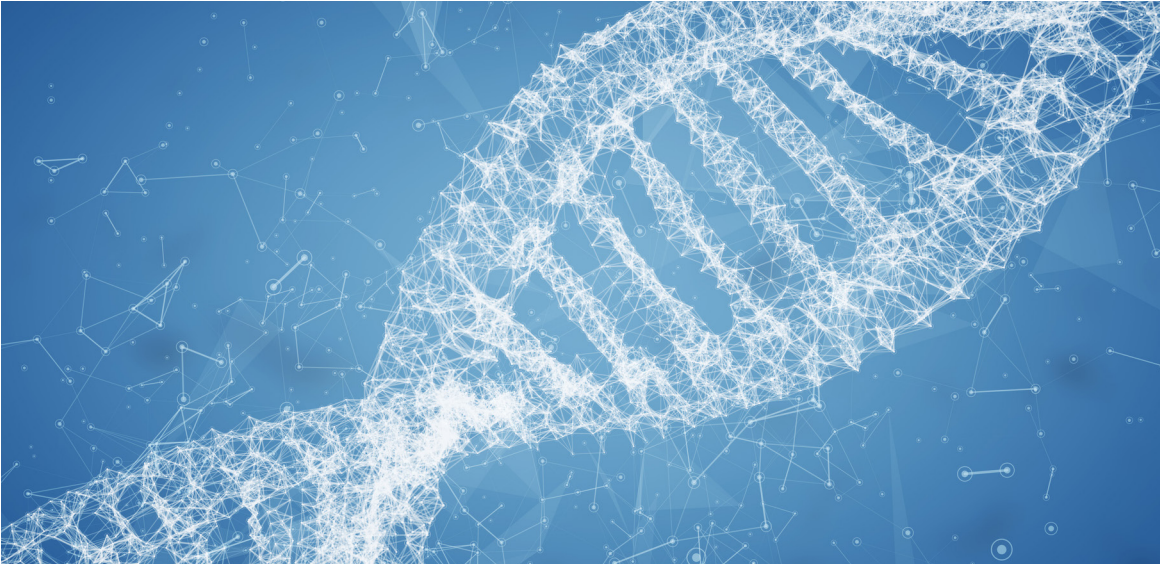
It is important that a sufficient number of technical replicates be tested on each eDNA sample to provide appropriate detection probability. The eDNA assay is pushed to the limits in detecting DNA at very low concentrations in environmental samples. As such, it is possible to have cases where individual qPCR replicates fail to generate a signal, or there is stochastic generation of a signal that is not dependable. It is recommended that 8 eDNA assay technical replicates be tested on each sample to reduce binomial error and provide statistical confidence in the assay result^{2,4}.

INTERPRETATION OF ENVIRONMENTAL DNA RESULTS

All samples and technical replicates are considered when interpreting eDNA results to determine detection, or non-detection of the target species. Given that individual technical replicates may or may not generate a signal, it is not appropriate to use the result of a single replicate in isolation to make a decision on the presence of the target organism.

When interpreting results, it is recommended that a positive species detection include a review of additional factors including sensitivity of the test, expected DNA shedding rates and environmental conditions at the time of sampling.

⁴Veldhoen, N.; Hobbs, J.; Ikononou, G.; Hii, M.; Lesperance, M.; Helbing, C.C. 2016. Implementation of Novel Design Features for qPCR-Based eDNA Assessment. PLoS ONE 11(11):1-23.



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